

TrES-2b

R-1534

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_130805 · created 2026-08-01T18:29:51+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456509.77881 +/- 0.00024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.01 +/- 0.041
Transit depth (Rp/Rs)^2	0.011 +/- 0.086 [%]
Orbital Inclination (inc)	79.56 +/- 2.83
Airmass coefficient 1 (a1)	0.99 +/- 0.034
Airmass coefficient 2 (a2)	0.013 +/- 0.082
Scatter in the residuals of the lightcurve fit is	3.49 %
Best Comparison Star	#1 - [399, 186]
Optimal Aperture	2.81
Optimal Annulus	10.25
Transit Duration (day)	0.008 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.01 $\pm$ 0.041 vs 0.1254 (deviation 2.81 σ)
Duration fit vs published	0.008 d vs 0.0746 d (deviation 3.03 σ)
Mid-time O-C	-0.4 min
Depth SNR	0.2
Residual scatter	3.49 %

Light curve

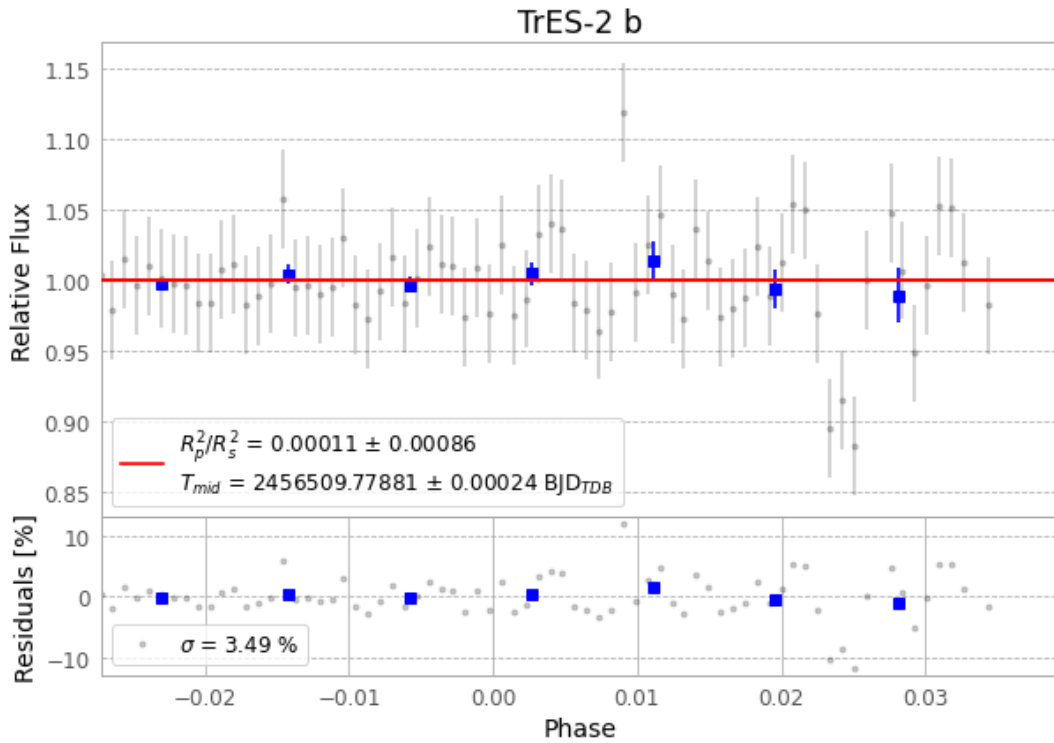


Figure 1 — FinalLightCurve_TrES-2 b_2013-08-04.png (SHA-256 e8a08f5904bdb19d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [393, 251], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b25559e3b6623212...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

80 FITS frames (53 MB), TRES-2130805050015.FITS → TRES-2130805085712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-130805.log (6cc041e6a637...), FinalLightCurve_TrES-2 b_2013-08-04.png (e8a08f5904bd...), FinalLightCurve_TrES-2 b_2013-08-04.csv (b6e1f90ef405...)

Compute resources

Wall time 355.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 18:29Z.